SUPPLEMENTARY MATERIALS

SURVIVAL AND INVASION DYNAMICS IN CELL POPULATIONS: AN ANALYTICAL FRAMEWORK FOR THRESHOLD BEHAVIOUR IN NONLINEAR AGE-STRUCTURED MODELS

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SM1. Model summary and reduced moment system. This document provides supplementary details for the age-structured model of proliferating cell populations analysed in the main text. The model tracks the density of cells, $u(a, x, t)$, of age a (time since last division) at position x and time t. The dynamics are governed by the transport-renewal equation with diffusion

$$\partial_t u + \partial_a u = \kappa \Delta u - [\mu(a, P) + \beta(a, P)]\, u, \tag{SM1.1}$$

$$u(0, x, t) = 2 \int_0^\infty \beta(a, P) u(a, x, t)\, \mathrm{d}a, \tag{SM1.2}$$

where $P(x, t) = \int_0^\infty u(a, x, t)\, \mathrm{d}a$ is the total population density.

We analyse different model cases (Table SM1): constant rates (age-homogeneous), exponential decay (early division), and gamma-type profiles (maturation delay).
These forms are chosen for analytical tractability and biological relevance. Each case is studied in both the non-spatial ($\kappa = 0$) and spatial ($\kappa > 0$) settings.

TABLE SM1
Model cases studied in both non-spatial and spatial settings

Case	$\mu(a, P)$	$\beta(a, P)$
1	μ	$\beta(1 - P)$
1b	μP	$\beta(1 - P)$
2	μ	$\beta e^{-\alpha a}(1 - P)$
3	μ	$\beta a e^{-\alpha a}(1 - P)$
4	μP	$\beta a e^{-\alpha a}(1 - P)$
5	$(\mu - \gamma a e^{-\alpha a})P$	$\beta a e^{-\alpha a}(1 - P)$

Generalised moment system. We derive an integro–differential reduction of the age-structured system (SM1.1)–(SM1.2). We introduce age-weighted moments that mirror the form of division rates.

For any fixed decay parameter $\alpha > 0$, the reduced system is formulated in terms of the following quantities,

$$P(x, t) = \int_0^\infty u(a, x, t)\, \mathrm{d}a, \tag{SM1.3}$$

$$C(x, t; \alpha) = \int_0^\infty e^{-\alpha a} u(a, x, t)\, \mathrm{d}a, \tag{SM1.4}$$

$$D(x, t; \alpha) = \int_0^\infty a e^{-\alpha a} u(a, x, t)\, \mathrm{d}a. \tag{SM1.5}$$

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The system (P, C, D) is not closed, and its evolution involves higher moments,

$$I_0(t,x;\alpha) = \int_0^\infty e^{-2\alpha a} u(a,x,t)\,\mathrm{d}a, \tag{SM1.6}$$

$$I_1(t,x;\alpha) = \int_0^\infty a e^{-2\alpha a} u(a,x,t)\,\mathrm{d}a, \tag{SM1.7}$$

$$I_2(t,x;\alpha) = \int_0^\infty a^2 e^{-2\alpha a} u(a,x,t)\,\mathrm{d}a. \tag{SM1.8}$$

The birth term (birth boundary condition) is case-dependent and involves these moments,

$$B(x,t) = u(0,x,t) = \begin{cases} 2\beta(1-P)P, & \text{Case 1,} \\ 2\beta(1-P)C, & \text{Case 2,} \\ 2\beta(1-P)D, & \text{Cases 3–5.} \end{cases} \tag{SM1.9}$$

The resulting generalised integro–differential system is

$$\partial_t P = \kappa\,\Delta P + B - \int_0^\infty [\mu(a,P(x,t)) + \beta(a,P(x,t))]\,u(a,x,t)\,\mathrm{d}a, \tag{SM1.10}$$

$$\partial_t C = \kappa\,\Delta C + B - \alpha C - \int_0^\infty [\mu(a,P(x,t)) + \beta(a,P(x,t))] e^{-\alpha a} u(a,x,t)\,\mathrm{d}a, \tag{SM1.11}$$

$$\partial_t D = \kappa\,\Delta D + C - \alpha D - \int_0^\infty [\mu(a,P(x,t)) + \beta(a,P(x,t))] a e^{-\alpha a} u(a,x,t)\,\mathrm{d}a, \tag{SM1.12}$$

with $B(x,t)$ (SM1.9) and higher moments $I_k(x,t;\alpha) = \int_0^\infty a^k e^{-2\alpha a} u(a,x,t)\mathrm{d}a$. The integral terms are evaluated using the defined moments and the specific forms of $\mu(a,P)$ and $\beta(a,P)$ for each model case. In general, age-homogeneous rates (constant rates) yield exact closure in $P(t)$; exponential decay rates (early reproduction bias) require bounds on P, C, I_0; gamma-type rates (maturation delay) require bounds on P, C, D, I_1, I_2.

SM2. Boundedness of age-weighted moments.

Proposition SM2.1 (Boundedness in the non-spatial case). *Suppose $\kappa = 0$ and $P(0) \le 1$. Then for all $t \ge 0$:*

(i) $P(t) \le 1$,
(ii) The moments $C(t;\alpha)$, $D(t;\alpha)$, and $I_k(t;\alpha)$ remain bounded.

Proof. **(i) Population bound.** The total population satisfies

$$\frac{\mathrm{d}P}{\mathrm{d}t} = B(t) - \int_0^\infty [\mu(a,P) + \beta(a,P)] u(a,t)\mathrm{d}a,$$

where $B(t) = 2(1-P)\int_0^\infty \beta_0(a) u(a,t)\mathrm{d}a$.

Whenever $P(t) \ge 1$, we have $B(t) \le 0$ whilst the loss term remains non-negative. Thus $\mathrm{d}P/\mathrm{d}t \le 0$ in any region where $P \ge 1$. Since $P(0) \le 1$, it follows that $P(t) \le 1$ for all $t \ge 0$.

(ii) Moment bounds. Since $e^{-\alpha a} < 1$ for all $a > 0$,

$$C(t;\alpha) = \int_0^\infty e^{-\alpha a} u(a,t)\mathrm{d}a < P(t) \le 1.$$

The function $a e^{-\alpha a}$ attains its maximum $1/(\alpha e)$ at $a = 1/\alpha$, giving

$$D(t;\alpha) = \int_0^\infty a e^{-\alpha a} u(a,t)\mathrm{d}a \le \frac{P(t)}{\alpha e} \le \frac{1}{\alpha e}.$$

Similarly, $a^k e^{-2\alpha a}$ attains its maximum $(k/(2\alpha e))^k$ at $a = k/(2\alpha)$, hence

$$I_k(t;\alpha) = \int_0^\infty a^k e^{-2\alpha a} u(a,t)\mathrm{d}a \le \left(\frac{k}{2\alpha e}\right)^k P(t) \le \left(\frac{k}{2\alpha e}\right)^k.$$

These bounds ensure all moments remain finite. ∎

SM3. Moment-based derivations of survival thresholds.

SM3.1. Fully closed population dynamics. When division and death rates depend only on total density, the population is age-homogeneous. All cells share the same division and death rates, and the moment system closes exactly, as established by Lemma 1 in the main text.

Case 1. Consider the rates

$$\mu(a,P)=\mu, \qquad \beta(a,P)=\beta(1-P), \tag{SM3.1}$$

with constants $\mu,\beta>0$. This models a population where division is subject to contact inhibition, while cell death occurs at a constant rate independent of local conditions. The total population $P(t)$ satisfies the logistic equation,

$$P'(t)=(\beta-\mu)\,P(t)\left(1-\frac{P(t)}{K}\right), \quad \text{with } K=\frac{\beta-\mu}{\beta}.$$

Population persistence requires $\beta>\mu$. When this holds, the unique positive steady state defines the critical threshold

$$P_c=1-\frac{\mu}{\beta}. \tag{SM3.2}$$

SM3.2. Exponential decay in division potential. This models a division rate that declines exponentially with age, $\beta(a,P)\propto e^{-\alpha a}$. This is a reasonable approximation if the population reproduces at sufficiently young ages.

Case 2. Consider the rates

$$\mu(a,P)=\mu, \qquad \beta(a,P)=\beta e^{-\alpha a}(1-P), \tag{SM3.3}$$

with constants $\mu,\beta,\alpha>0$. Division potential is highest immediately after birth and decays exponentially thereafter. The system (SM1.1)–(SM1.2) reduces to

$$\begin{aligned}P'(t)&=-\mu P(t)+\beta(1-P(t))C(t),\\ C'(t)&=-(\mu+\alpha)C(t)+\beta(1-P(t))\left(2C(t)-I_0(t)\right),\end{aligned}$$

where $C(t)=\int_0^\infty e^{-\alpha a}u\,\mathrm{d}a$ and $I_0(t)=\int_0^\infty e^{-2\alpha a}u\,\mathrm{d}a$.

At steady state,

$$\bar{C}=\frac{\mu\,\bar{P}}{\beta(1-\bar{P})}, \quad \bar{I}_0=2\bar{C}-\frac{(\mu+\alpha)\,\bar{C}}{\beta(1-\bar{P})}, \quad \text{for } \bar{P}\neq 1.$$

The non-negativity constraint $\bar{I}_0\geq 0$ implies

$$2\bar{C}-\frac{(\mu+\alpha)\,\bar{C}}{\beta(1-\bar{P})}\geq 0.$$

Assuming $\bar{C}\neq 0$ and $\bar{P}\neq 1$, this simplifies to $2\beta(1-\bar{P})\geq\mu+\alpha$. Therefore, population persistence requires $2\beta>\mu+\alpha$, and any positive steady state satisfies $\bar{P}<P_c$, with

$$P_c=1-\frac{\mu+\alpha}{2\beta}. \tag{SM3.4}$$

If $2\beta\leq\mu+\alpha$, only the extinction equilibrium exists.

SM3.3. Gamma-type age dependence. The division rate follows a gamma-type profile $\beta(a,P)\propto ae^{-\alpha a}$, capturing maturation delays before division. We consider three cases: constant mortality (Case 3), density-dependent mortality (Case 4), and age- and density-dependent mortality (Case 5).

Case 3. Consider the rates

$$\mu(a,P)=\mu, \qquad \beta(a,P)=\beta a e^{-\alpha a}(1-P), \tag{SM3.5}$$

with constants $\mu,\beta,\alpha>0$. The system (SM1.1)–(SM1.2) reduces to

$$\begin{aligned} P'(t) &= -\mu P(t)+\beta(1-P(t))\,D(t),\\ C'(t) &= -(\mu+\alpha)\,C(t)+\beta(1-P(t))\,(2D(t)-I_1(t)),\\ D'(t) &= -(\mu+\alpha)\,D(t)+C(t)-\beta(1-P(t))\,I_2(t), \end{aligned} \tag{SM3.6}$$

with

$$C(t;\alpha)=\int_0^\infty e^{-\alpha a}u(a,t)\,\mathrm{d}a, \qquad D(t;\alpha)=\int_0^\infty a e^{-\alpha a}u(a,t)\,\mathrm{d}a,$$

and higher moments

$$I_1(t;\alpha)=\int_0^\infty a e^{-2\alpha a}u(a,t)\,\mathrm{d}a, \qquad I_2(t;\alpha)=\int_0^\infty a^2 e^{-2\alpha a}u(a,t)\,\mathrm{d}a.$$

At steady state, assuming $\bar{D}\neq 0$, we obtain

$$\bar{D}=\frac{\mu\,\bar{P}}{\beta(1-\bar{P})}, \qquad \bar{I}_1=2\bar{D}-\frac{(\mu+\alpha)\,\bar{C}}{\beta(1-\bar{P})}, \qquad \bar{I}_2=\frac{\bar{C}-(\mu+\alpha)\,\bar{D}}{\beta(1-\bar{P})}, \quad \text{for } \bar{P}\neq 1.$$

Given $\bar{I}_1\geq 0$, we have

$$\bar{C}\leq\frac{2\mu\,\bar{P}}{\mu+\alpha}.$$

Substituting into $\bar{I}_2\geq 0$ yields

$$\bar{C}\,\beta(1-\bar{P})-(\mu+\alpha)\mu\,\bar{P}\geq 0 \quad\Longrightarrow\quad \bar{P}\leq\frac{\bar{C}\,\beta(1-\bar{P})}{(\mu+\alpha)\mu}\leq\frac{2\mu\,\bar{P}\,\beta(1-\bar{P})}{\mu(\mu+\alpha)^2}.$$

Assuming $\bar{P}\neq 0$, this gives

$$\bar{P}\leq 1-\frac{(\mu+\alpha)^2}{2\beta}.$$

For $\bar{P}$ to be positive, we require $2\beta>(\mu+\alpha)^2$. When this condition holds,

$$P_c=1-\frac{(\mu+\alpha)^2}{2\beta} \tag{SM3.7}$$

provides an upper bound on steady-state populations.

Case 4. Consider that death rate increases proportionally with total population,

$$\mu(a,P)=\mu P, \qquad \beta(a,P)=\beta a e^{-\alpha a}(1-P), \tag{SM3.8}$$

with constants $\mu,\beta,\alpha>0$. The moment hierarchy includes P, C, D, I_1, and I_2,

$$\begin{aligned} P'(t) &= -\mu P(t)^2+\beta(1-P(t))D(t),\\ C'(t) &= -(\mu P(t)+\alpha)C(t)+\beta(1-P(t))\,(2D(t)-I_1(t)),\\ D'(t) &= -(\mu P(t)+\alpha)D(t)+C(t)-\beta(1-P(t))I_2(t). \end{aligned} \tag{SM3.9}$$

At steady state,

$$\bar{D}=\frac{\mu\bar{P}^2}{\beta(1-\bar{P})}, \quad \bar{I}_1=2\bar{D}-\frac{\bar{C}(\alpha+\mu\bar{P})}{\beta(1-\bar{P})} \quad \bar{I}_2=\frac{\bar{C}-\bar{D}(\alpha+\mu\bar{P})}{\beta(1-\bar{P})}, \text{ for } \bar{P}\neq 1.$$

Given $\bar{I}_1\geq 0$,

$$\bar{C}\leq\frac{2\bar{D}\beta(1-\bar{P})}{\mu\bar{P}+\alpha}.$$

Similarly, $\bar{I}_2 \geq 0$ implies

$$\bar{D}(\alpha + \mu\bar{P}) \leq \bar{C}.$$

Combining these inequalities gives upper and lower bounds on $\bar{C}$, which lead to

$$(\mu\bar{P} + \alpha)^2 \leq 2\beta(1 - \bar{P}). \tag{SM3.10}$$

The above inequality simplifies to

$$\bar{P} \leq \frac{-(\alpha\mu + \beta) + \sqrt{(\alpha\mu + \beta)^2 - \mu^2(\alpha^2 - 2\beta)}}{\mu^2}, \quad \text{with} \quad 2\beta > \alpha^2. \tag{SM3.11}$$

Population survival thus requires $2\beta > \alpha^2$. When this holds,

$$P_c = \frac{-(\alpha\mu + \beta) + \sqrt{(\alpha\mu + \beta)^2 - \mu^2(\alpha^2 - 2\beta)}}{\mu^2}. \tag{SM3.12}$$

Case 5. Consider that both death and division rates depend explicitly on cell age,

$$\mu(a, P) = (\mu - \gamma\, a e^{-\alpha a})\, P, \qquad \beta(a, P) = \beta\, a e^{-\alpha a}(1 - P), \tag{SM3.13}$$

with $\mu, \gamma, \beta, \alpha > 0$. We require $\mu > \gamma/(\alpha e)$ to ensure positive death rates. This models a scenario where the death rate is initially elevated for newly divided cells, decreases during a maturation phase, and rises again due to senescence. The moment hierarchy becomes

$$\begin{aligned} P'(t) &= -\mu P(t)^2 + \gamma P(t) D(t) + \beta(1 - P(t)) D(t), \\ C'(t) &= -(\mu P(t) + \alpha) C(t) + \gamma P(t) I_1(t) + \beta(1 - P(t))\,(2D(t) - I_1(t))\,, \\ D'(t) &= -(\mu P(t) + \alpha) D(t) + C(t) + \gamma P(t) I_2(t) - \beta(1 - P(t)) I_2(t). \end{aligned} \tag{SM3.14}$$

At steady state with $\bar{P} \neq 1$ and $\gamma\bar{P} \neq \beta(1 - \bar{P})$ (ensuring the moment expressions are well-defined),

$$\bar{D} = \frac{\mu\bar{P}^2}{\gamma\bar{P} + \beta(1 - \bar{P})}, \quad \bar{I}_1 = \frac{(\mu\bar{P} + \alpha)\bar{C} - 2\beta(1 - \bar{P})\bar{D}}{\gamma\bar{P} - \beta(1 - \bar{P})}, \quad \bar{I}_2 = \frac{(\mu\bar{P} + \alpha)\bar{D} - \bar{C}}{\gamma\bar{P} - \beta(1 - \bar{P})}.$$

For non-negative moments $\bar{I}_1 \geq 0$ and $\bar{I}_2 \geq 0$, both numerator and denominator must have the same sign. The condition $\gamma\bar{P} \neq \beta(1 - \bar{P})$ ensures the denominators are non-zero.

We analyse first the case where both the numerator and denominator are negative in the expressions for $\bar{I}_1$ and $\bar{I}_2$. From $\bar{I}_1 \geq 0$ we obtain

$$\bar{C} \leq \frac{2\beta(1 - \bar{P})}{\mu\bar{P} + \alpha}\bar{D}.$$

Then from $\bar{I}_2 \geq 0$ we obtain

$$\bar{C} \geq (\mu\bar{P} + \alpha)\bar{D}.$$

Combining these inequalities gives the constraint

$$(\mu\bar{P} + \alpha)^2 \leq 2\beta(1 - \bar{P}). \tag{SM3.15}$$

The function $f(P) = (\mu P + \alpha)^2 + 2\beta P - 2\beta$ is strictly increasing for $P \geq 0$, since

$$f'(P) = 2\mu(\mu P + \alpha) + 2\beta > 0.$$

Therefore, the inequality $f(\bar{P}) \leq 0$ is equivalent to $\bar{P} \leq P_c$, where P_c is the unique positive root of $f(P) = 0$,

$$P_c = \frac{-(\alpha\mu + \beta) + \sqrt{(\alpha\mu + \beta)^2 - \mu^2(\alpha^2 - 2\beta)}}{\mu^2}. \tag{SM3.16}$$

The condition $2\beta > \alpha^2$ ensures the discriminant is positive and $P_c > 0$. From the denominator we have $\bar{P} < \beta/(\gamma+\beta)$ (strict inequality since $\gamma\bar{P} \neq \beta(1-\bar{P})$). The upper bound on the positive steady state becomes

$$\bar{P} < \min\left(P_c,\ \beta/(\gamma+\beta)\right).$$

When both numerator and denominator are positive, the analogous derivation yields $(\mu\bar{P}+\alpha)^2 \geq 2\beta(1-\bar{P})$, which combined with $\bar{P} > \beta/(\gamma+\beta)$ provides a lower bound on feasible steady states. However, for survival threshold analysis, the first case provides the relevant upper bound. The survival condition $2\beta > \alpha^2$ ensures positive steady states exist with population bounded by the constraints above.

The parameter γ determines how strongly the death rate varies with cell age. In the limit $\gamma \to 0$, the model reduces to Case 4 with purely density-dependent death, and $\beta/(\gamma+\beta) \to 1$, leaving P_c as the sole bound. As γ increases, the constraint $\mu > \gamma/(\alpha e)$ forces proportionally larger baseline death rates μ, whilst $\beta/(\gamma+\beta) \to 0$ drives the equilibrium toward extinction. Although positive steady states persist when $2\beta > \alpha^2$, large γ yields populations barely above zero. For typical biological scenarios, γ remains small relative to β, ensuring P_c provides the relevant upper bound.

SM4. Minimal wave speed derivations. We present the full derivation for Case 5, which is the most general. Case 1 is derived in the main text. Cases 2–4 follow by simplification (replacing μ with μP, setting $\gamma = 0$, or removing age-dependent terms).

Consider the following division and death rates,

$$\mu(a,P) = (\mu - \gamma a e^{-\alpha a})\,P, \qquad \beta(a,P) = \beta a e^{-\alpha a}(1-P). \tag{SM4.1}$$

The survival condition from the non-spatial analysis is $2\beta > \alpha^2$, which we assume holds. We further require $\mu > \gamma/(\alpha e)$ to ensure $\mu(a,P) > 0$ for all $a \geq 0$.

We derive the minimal speed $c_{\min}$ via the dispersion relation arising from the full age-structured PDE. Starting from

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial a} = \kappa\frac{\partial^2 u}{\partial x^2} - [\mu(a,P) + \beta(a,P)]u, \tag{SM4.2}$$

with boundary condition

$$u(0,x,t) = 2\int_0^\infty \beta(a,P(x,t))u(a,x,t)\,\mathrm{d}a, \tag{SM4.3}$$

the travelling-wave ansatz $u(a,x,t) = \phi(a,z)$, $z = x - ct$ transforms the system into

$$-c\frac{\partial \phi}{\partial z} + \frac{\partial \phi}{\partial a} = \kappa\frac{\partial^2 \phi}{\partial z^2} - [(\mu - \gamma a e^{-\alpha a})U(z) + \beta a e^{-\alpha a}(1-U(z))]\phi, \tag{SM4.4}$$

where $U(z) = \int_0^\infty \phi(a,z)\,\mathrm{d}a$.

Linearising at the leading edge where $U \to 0$, the death term $(\mu - \gamma a e^{-\alpha a})U \to 0$ and $(1-U) \to 1$, giving

$$-c\frac{\partial \phi}{\partial z} + \frac{\partial \phi}{\partial a} = \kappa\frac{\partial^2 \phi}{\partial z^2} - \beta a e^{-\alpha a}\phi + O(U). \tag{SM4.5}$$

The boundary condition linearises to

$$\phi(0,z) = 2\beta\int_0^\infty a e^{-\alpha a}\phi(a,z)\,\mathrm{d}a + O(U). \tag{SM4.6}$$

We assume separability $\phi(a,z) = A(a)e^{-\lambda z}$ and solve the resulting age-profile equation. Here λ is the rate of spatial decay. This yields the dispersion relation

$$1 = 2\beta\int_0^\infty a e^{-\alpha a}\exp\left[-(c\lambda - \kappa\lambda^2)a - \frac{\beta}{\alpha^2}(1 - e^{-\alpha a}(1+\alpha a))\right]\mathrm{d}a. \tag{SM4.7}$$

The minimal speed $c_{\min}$ is the smallest $c > 0$ for which there exists $\lambda > 0$ satisfying equation (SM4.7).

We now derive the upper bound c_{lin} from the moment system. Using the reduction framework of Section 2 in the main text and the travelling-wave ansatz $z = x - ct$, we have

$$\begin{aligned} -cU' &= \kappa U'' - \mu U^2 + \gamma UW + \beta(1-U)W, \\ -cV' &= \kappa V'' - (\mu U + \alpha)V + \gamma U\tilde{I}_1 + \beta(1-U)\left(2W - \tilde{I}_1\right), \\ -cW' &= \kappa W'' - (\mu U + \alpha)W + V + \gamma U\tilde{I}_2 - \beta(1-U)\tilde{I}_2. \end{aligned} \tag{SM4.8}$$

where $P(x,t) = U(z)$, $C(x,t) = V(z)$, $D(x,t) = W(z)$, and $I_k(x,t) = \tilde{I}(z)$. Converting to a first-order system with $Y_1 = U$, $Y_2 = U'$, $Y_3 = V$, $Y_4 = V'$, $Y_5 = W$, $Y_6 = W'$, the linearisation at $(Y_1, \ldots, Y_6) = (0, \ldots, 0)$ yields

$$\mathbf{Y}' = \mathbf{J}\,\mathbf{Y}, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\frac{c}{\kappa} & 0 & 0 & -\frac{\beta}{\kappa} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{\alpha}{\kappa} & -\frac{c}{\kappa} & -\frac{2\beta}{\kappa} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{1}{\kappa} & 0 & \frac{\alpha}{\kappa} & -\frac{c}{\kappa} \end{pmatrix}.$$

The eigenvalues are

$$\lambda_1 = 0, \quad \lambda_2 = -\frac{c}{\kappa},$$

$$\lambda_{3,4} = \frac{-c \pm \sqrt{c^2 + 4(\alpha - \sqrt{2\beta})\kappa}}{2\kappa},$$

$$\lambda_{5,6} = \frac{-c \pm \sqrt{c^2 + 4(\alpha + \sqrt{2\beta})\kappa}}{2\kappa}.$$

We seek biologically realistic solutions, and hence the eigenvalues must be real. The critical constraint comes from λ_3, which has positive real part unless $c^2 + 4(\alpha - \sqrt{2\beta})\kappa \le 0$. This requires

$$c \ge 2\sqrt{\kappa(\sqrt{2\beta} - \alpha)},$$

yielding the upper bound $c_{\text{lin}} = 2\sqrt{\kappa(\sqrt{2\beta} - \alpha)}$, valid when $\sqrt{2\beta} > \alpha$.

The inequality $c_{\min} \le c_{\text{lin}}$ reflects the fact that the moment system linearisation assumes all age classes vanish uniformly at the wave front, neglecting the natural age structure where younger cells dominate the leading edge. The dispersion relation preserves this age-dependent decay structure at the leading edge, yielding a sharper (and typically slower) minimal speed bound. Both approaches give the same threshold condition for invasion: $\sqrt{2\beta} > \alpha$, equivalent to the non-spatial survival condition $2\beta > \alpha^2$.

SM5. Numerical methods. The system (SM1.1)–(SM1.2) combines three distinct processes: advective transport in the age variable, diffusive spread in space, and a non-local renewal condition at $a = 0$ that links all ages through cell division. Because these processes act on different variables and timescales, advancing them simultaneously introduces numerical errors. The diffusion operator acts locally in space, coupling only neighbouring points, whereas the renewal term introduces a global coupling across the age domain. Standard reaction–diffusion schemes fail to maintain the non-negativity of $u(a,x,t)$ and to balance the total renewal flux. Additional numerical challenges arise from the unbounded age domain and from stiffness when the combined loss rate $\mu(a,P) + \beta(a,P)$ changes rapidly with age or density.

We address these challenges with an operator-splitting scheme that advances age transport, reaction, and diffusion sequentially, combined with a predictor–corrector integration that evaluates rates at both the current and predicted states to achieve second-order accuracy. The renewal boundary condition is computed from the corrected interior solution using averaged division rates. This ensures consistent coupling between the non-local birth integral and the interior dynamics while preserving mass balance and non-negativity throughout the domain.

Discretisation. The computational domain is $x \in [-L, L]$, $a \in [0, a_{\max}]$, and $t \in [0, T]$. We discretise the solution $u(a, x, t)$ on a uniform grid: $a_j = j\,\Delta a$ for age, $x_i = i\,\Delta x$ for space, and $t_n = n\,\Delta t$ for time. Here $u^n_{j,i}$ denotes the numerical approximation to $u(a_j, x_i, t_n)$.

For the age variable, the discrete transport operator is constructed along the exact characteristics of the hyperbolic part, $\partial_t u + \partial_a u = 0$, whose solutions satisfy $u(a, t) = u(a - t, 0)$. Setting $\Delta t = \Delta a$ ensures that

$$u^{n+1}_{j+1,i} = u^n_{j,i},$$

meaning that each cohort of cells ages exactly one bin per time step. This choice guarantees positivity and eliminates artificial smoothing in the age coordinate. It also couples the time step to the age grid, a mild but necessary constraint for accuracy and stability. Stability is maintained by ensuring that $\Delta a = \Delta t$ is sufficiently small relative to the reaction timescales.

For spatial diffusion, we employ the Crank–Nicolson scheme, treating the diffusion operator implicitly via a centred three-point stencil. At each time step, the update for age slice j satisfies

$$\frac{u^{n+1}_{j,i} - u^n_{j,i}}{\Delta t} = \frac{\kappa}{2}\,\frac{u^{n+1}_{j,i-1} - 2u^{n+1}_{j,i} + u^{n+1}_{j,i+1}}{(\Delta x)^2} + \frac{\kappa}{2}\,\frac{u^n_{j,i-1} - 2u^n_{j,i} + u^n_{j,i+1}}{(\Delta x)^2},$$

which is second-order accurate in both space and time for the diffusive subproblem and unconditionally stable. We impose zero-flux boundary conditions at $x = \pm L$: at the left boundary, $u_{-1,j} = u_{1,j}$; at the right boundary, $u_{n_x+1,j} = u_{n_x-1,j}$. Thus $\partial_x u = 0$ at the edges.

Operator splitting and nonlinear coupling. Each full step $t_n \to t_{n+1}$ proceeds through an operator-split sequence (age transport, reaction, diffusion) embedded within a predictor–corrector framework that makes the scheme second order in time.

In the predictor step, we compute rates $\mu(a, P^n)$ and $\beta(a, P^n)$ from the current population density $P^n = \int_0^{a_{\max}} u^n(a, x)\,\mathrm{d}a$, apply the explicit reaction update, and then advance diffusion via Crank–Nicolson to obtain the predicted solution u^{pred}. We recompute $P^{\text{pred}} = \int_0^{a_{\max}} u^{\text{pred}}(a, x)\,\mathrm{d}a$ and evaluate rates at this predicted state. The corrector step applies time-averaged rates $\bar{\mu} = \frac{1}{2}[\mu(a, P^n) + \mu(a, P^{\text{pred}})]$ and $\bar{\beta} = \frac{1}{2}[\beta(a, P^n) + \beta(a, P^{\text{pred}})]$ to the original u^n, again sequencing reaction then diffusion, yielding the corrected solution u^{corr}. This split predictor–corrector scheme achieves $\mathcal{O}(\Delta t^2)$ temporal accuracy. The renewal boundary condition at $a = 0$ is computed from the corrected interior solution using the averaged division rates. This ensures consistent boundary–interior coupling.

Age transport and renewal. The age variable evolves along the exact characteristics of the transport equation $\partial_t u + \partial_a u = 0$. With $\Delta t = \Delta a$, each cohort advances deterministically via $u^{n+1}_{j+1,i} = u^{n+1}_{j,i}$ for $j \geq 1$. The boundary at $a = 0$ is determined by the discrete renewal condition

$$u^{n+1}_{0,i} = 2 \sum_{j=0}^{N_a - 1} \bar{\beta}(a_j, P_i)\, u^{\text{corr}}_{j,i}\, w_j,$$

where w_j are Simpson quadrature weights, $\bar{\beta}$ denotes the time-averaged division rate from the predictor–corrector scheme, and u^{corr} is the corrected interior solution. This maintains the integral balance and preserves non-negativity provided $\beta \geq 0$.

The unbounded age domain $[0, \infty)$ is truncated dynamically at $a_{\max}(t)$. At each step we compute the tail fraction

$$\eta(t) = \frac{\int_{-L}^{L} u(a_{\max}, x, t)\,\mathrm{d}x\,\Delta a}{\int_0^{a_{\max}} \int_{-L}^{L} u(a, x, t)\,\mathrm{d}x\,\mathrm{d}a}.$$

If $\eta(t) > \eta_{\text{tol}} = 10^{-6}$, we extend the grid by one age bin; otherwise, the oldest cohort is removed. This adaptive procedure bounds the relative truncation error in total mass by $\mathcal{O}(\eta_{\text{tol}})$ while maintaining computational cost linear in the number of active age classes.

SM5.1. Numerical validation. We validate the scheme through three complementary tests. First, mass conservation: in non-spatial simulations across all five model cases, the relative mass drift remains below 10^{-6} after transients (Figure SM1). Second, travelling wave speeds: numerical estimates c_{est} converge

to theoretical predictions $c_{\min}$ across Cases 1–5, confirming the linearisation analysis and dispersion relations (Figure SM2). Third, grid refinement: L^2 errors in both $P(x)$ and $u(a,x)$ exhibit first-order convergence in Δa and Δx (Figure SM3), consistent with the characteristic marching in the age coordinate, which is formally first-order despite the higher-order treatment of diffusion and reaction terms. Together, these tests confirm the scheme faithfully reproduces the analytical framework whilst maintaining conservation laws and the predicted wave dynamics.

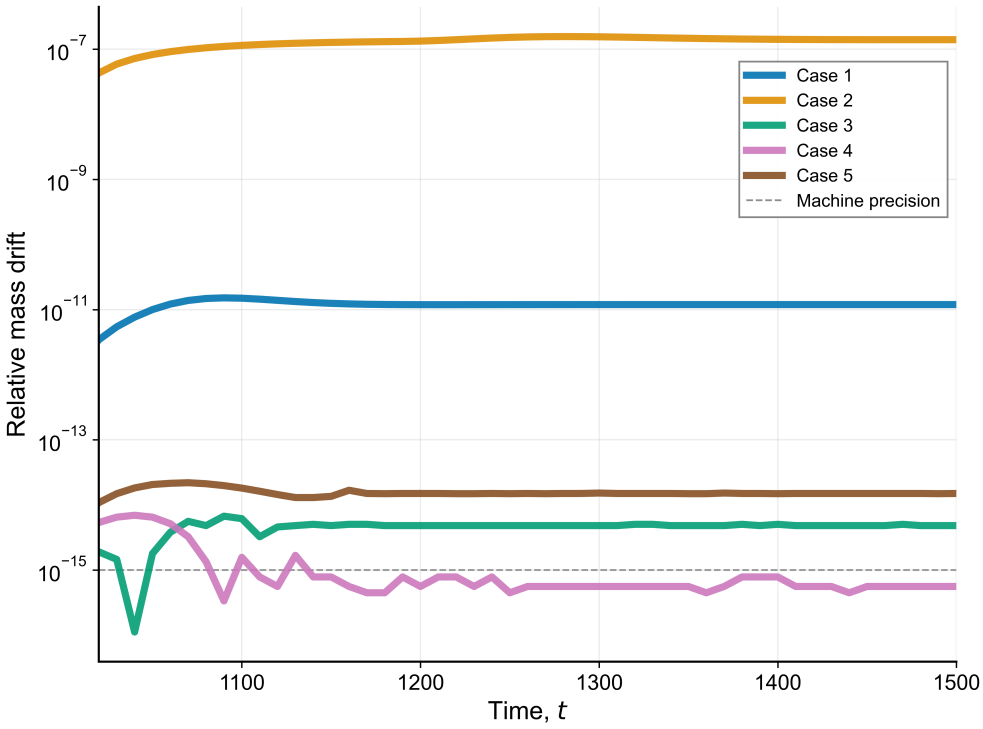

FIG. SM1. *Mass conservation validation for the non-spatial numerical scheme. Relative mass drift over time is shown for all model cases (1–5) on a logarithmic scale. Simulations use parameters* $T = 1500$, $\Delta a = 10^{-2}$, $\alpha = 0.01$, $\mu = 0.012$, $\beta = 0.065$, *and* $\gamma = 10^{-5}$. *The parameter* γ *applies only to Case 5. The initial condition is* $u(a,0) = e^{-10a^2}$. *All cases conserve total mass, with relative drift remaining below* 10^{-6}.

SM6. Parameter estimation from cell-cycle time data. Parameters for Case 3 (β, α, μ) were estimated from experimental intermitotic time distributions using a profile likelihood approach. The minimum cell-cycle time a_0 was fixed at the 10th percentile of each dataset to define the left support of the distribution.

For a grid of α values spanning a plausible range determined by the data quantiles (specifically, $\alpha \in [1/t_{50}, 1/t_{10}]$ where t_{10} and t_{50} are the 10th and 50th percentiles of the shifted data), we optimised (β, μ) by minimising the combined objective function

$$L(\beta, \mu \mid \alpha) = L_{\text{CDF}} + \lambda\, L_{\text{tail}} + P(\beta, \mu, \alpha),$$

where

- $L_{\text{CDF}} = \int_{a_0}^{a_{\max}} [F_{\text{model}}(a) - F_{\text{data}}(a)]^2 \, da$ measures the L^2 error between the model and empirical cumulative distribution functions (CDFs),
- $L_{\text{tail}} = \frac{1}{3} \sum_i [S_{\text{model}}(a_i) - S_{\text{data}}(a_i)]^2$ penalises deviations in survival probabilities at anchor points a_i corresponding to the 80th percentile, 90th percentile, and maximum observed time,
- $P(\beta, \mu, \alpha) = 10^6 \cdot \max(0, (\mu+\alpha)^2 - 2\beta)^2$ enforces the survival condition $2\beta > (\mu+\alpha)^2$ via a quadratic penalty.

The weight $\lambda = 0.5$ balances CDF accuracy with tail behavior. Results were insensitive to $\lambda \in [0.3, 0.7]$. The triplet (β, α, μ) minimising L across all α values was selected as the final estimate. Optimisation was

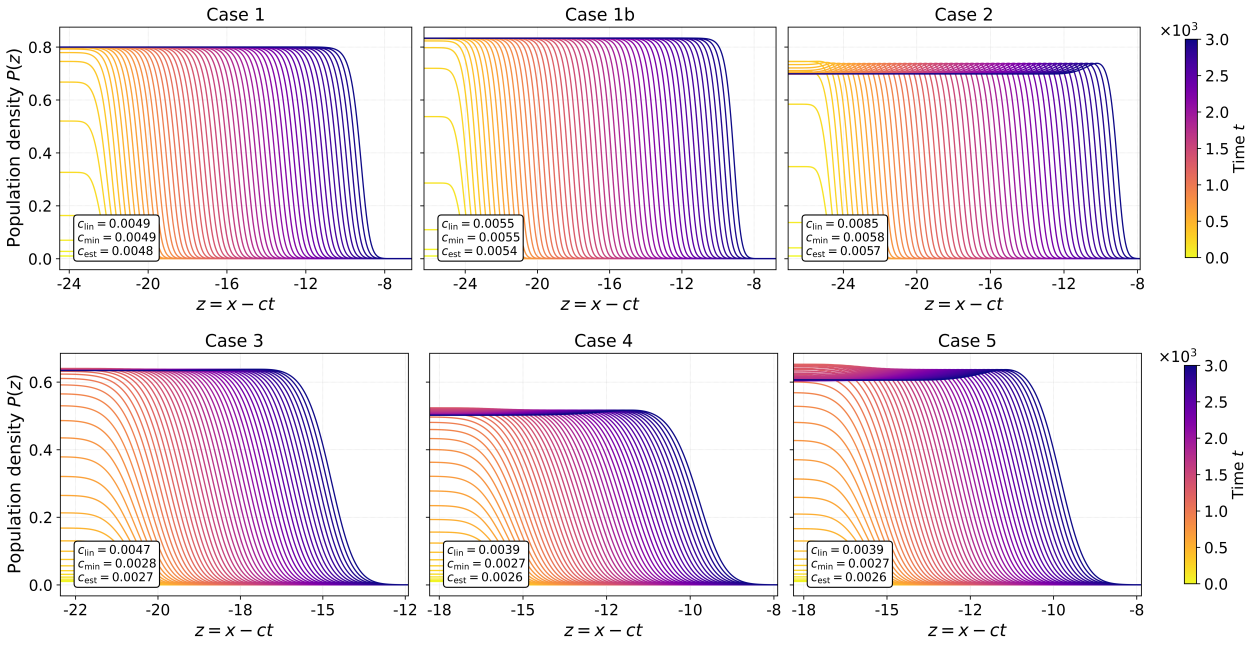

FIG. SM2. *Travelling-wave solutions for Cases1–5. Each panel shows the population density $P(z)$ in the moving frame $z = x - ct$. Numerical solutions converge to stable travelling waves with speeds matching theoretical predictions: for age-independent rates (Cases 1, 1b), $c_{\min} = c_{\lim}$; for models with maturation delay (Cases 2–5), invasion is slower, $c_{\min} < c_{\lim}$. In all cases, the estimated speed c_{est} agrees with the analytical $c_{\min}$ from the dispersion relation (5.9) in the main text. All simulations use domain $x \in [-10, 10]$, with parameters $\kappa = 3 \times 10^{-4}, \mu = 0.005, \alpha = 0.01, \gamma = 0.0001$. The parameter γ applies only to Case 5. Division rates are $\beta = 0.025$ (Cases 1, 1b), 0.0375 (Case 2), 0.0005625 (Case 3), and 0.00025 (Cases 4–5). The rates β were selected so that all cases satisfy the same relative distance above their survival condition, with $r = 2\beta/(\mu+\alpha) = 5$ (or the corresponding expression for each model). This ensures comparable invasion regimes across models. Initial condition: $u(a, x, 0) = 0.01\mathbb{1}_{x<-8}e^{-a}$.*

performed using L-BFGS-B with box constraints $\beta \in [10^{-3}, 2.0]$ and $\mu \in [10^{-4}, 1.0]$. Complete Python code for fitting and visualisation is available on GitHub at https://github.com/Stephanie-Abo/age-structured-invasion-dynamics.git.

Declarations

Authorship and contributorship. All authors have made substantial intellectual contributions to the study conception, execution, and design of the work. All authors have read and approved the final manuscript. Contributions were as follows: Conceptualisation: Stéphanie M. C. Abo, Ruth E. Baker; Methodology: Stéphanie M. C. Abo, Ruth E. Baker; Software: Stéphanie M. C. Abo; Formal analysis and investigation: Stéphanie M. C. Abo; Writing – original draft preparation: Stéphanie M. C. Abo; Writing – review and editing: Stéphanie M. C. Abo, Ruth E. Baker; Supervision: Ruth E. Baker; Funding acquisition: Ruth E. Baker.

Funding. Ruth E. Baker is supported by a grant from the Simons Foundation (MP-SIP-00001828).

Acknowledgements. We thank Christian A. Yates for sharing unpublished cell-cycle data originally collected by Richard L. Mort and Matthew J. Ford, and Rachel N. Leander for providing access to the DMSO, CHX, and ERL datasets available at https://github.com/rnleander/DDT_cell_cycle. We also thank Mike B. Giles for helpful discussions on the formulation and implementation of the numerical scheme.

Conflicts of interest. The authors declare there are no conflicts of interest.

Consent for publication All the authors approved the final version of the manuscript.

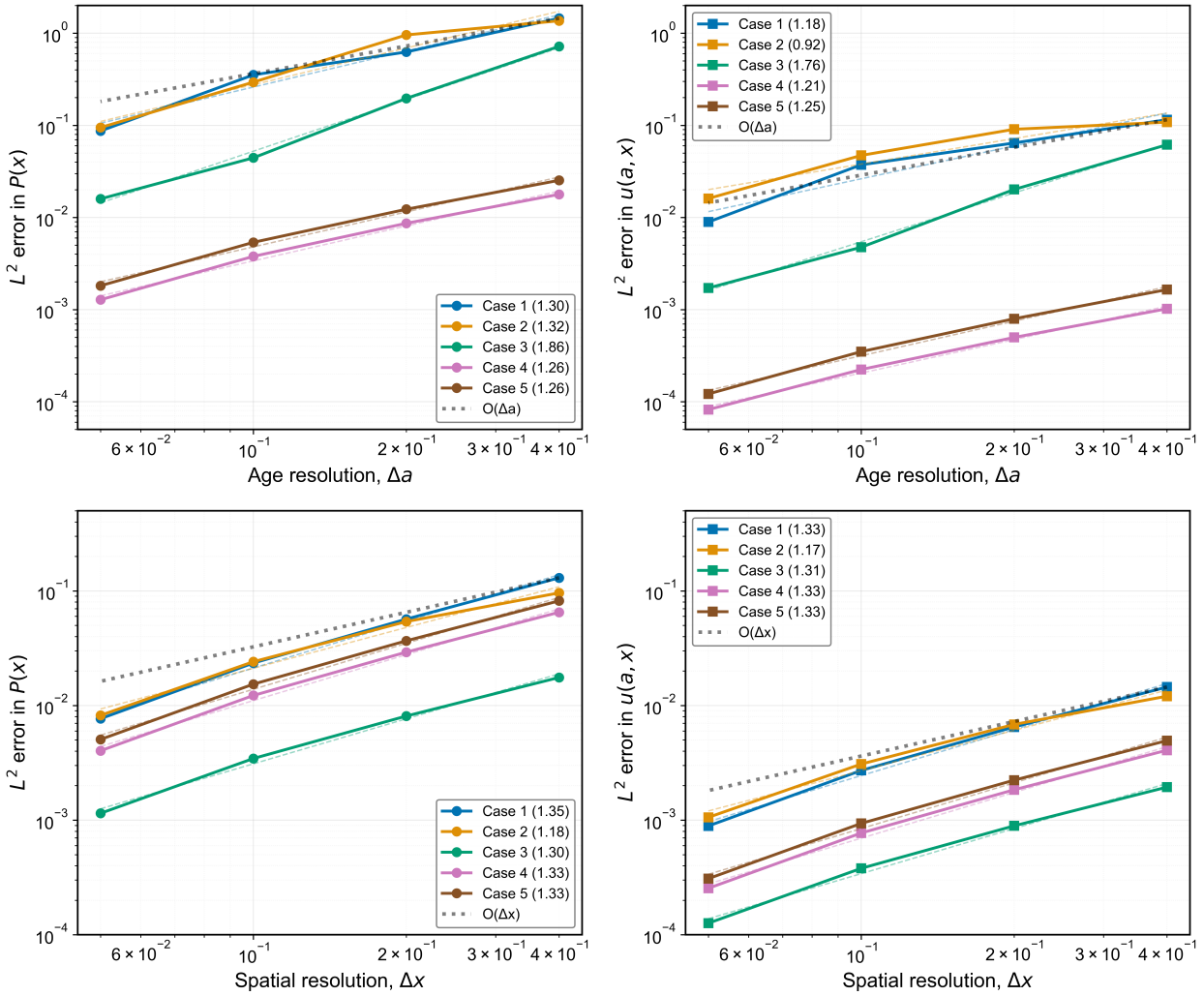

FIG. SM3. *Spatial and temporal convergence of the numerical scheme.* ***Top row****: L^2 error in total population $P(x)$ (left) and age distribution $u(a, x)$ (right) versus age-step size Δa (spatial resolution fixed at $\Delta x = 0.1$).* ***Bottom row****: L^2 error in $P(x)$ (left) and $u(a, x)$ (right) versus spatial-step size Δx (age resolution fixed at $\Delta a = 0.1$). All simulations use domain $x \in [-10, 10]$, with parameters $\kappa = 3 \times 10^{-4}, \mu = 0.005, \alpha = 0.01, \gamma = 0.0001$. The parameter γ applies only to Case 5. Division rates are $\beta = 0.025$ (Cases 1, 1b), 0.0375 (Case 2), 0.0005625 (Case 3), and 0.00025 (Cases 4–5). The rates β were selected so that all cases satisfy the same relative distance above their survival condition, with $r = 2\beta/(\mu + \alpha) = 5$ (or the corresponding expression for each model). This ensures comparable invasion regimes across models. Initial condition: $u(a, x, 0) = 0.01 \mathbb{1}_{x<-8} e^{-a}$. Errors are measured at steady wave profiles relative to finer reference grids ($\Delta a = \Delta x = 0.05$). Numbers in parentheses show convergence slopes from log–log regression; dotted lines mark first-order references $O(\Delta a)$ and $O(\Delta x)$. Age resolution gives near-first-order convergence (slopes $\approx$ 1.2–1.3, except Case 3 $\approx$ 1.8), while spatial resolution yields $\approx$ 1.3 across cases—consistent with coupling of first-order age transport with Crank–Nicolson diffusion.*

Code availability. All code used for the simulations and figure generation is publicly available on GitHub at https://github.com/Stephanie-Abo/age-structured-invasion-dynamics.git.